Impacts of H3K9me2 in Schizophrenia

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Introduction

- Schizophrenia (SCZ) is a chronic psychiatric disorder driven in part by epigenetic dysregulation in the brain
- H3K9me2, a repressive histone mark, is understudied yet suspected to alter gene expression in SCZ
- We investigated whether H3K9me2 modifications correlate with differential gene expression in the PFC (Brodmann areas 9 & 10) of SCZ patients

Data

- ChIP-seq (H3K9me2): 15 SCZ, 15 controls
  - PRJNA891668, Frontiers in Psychiatry (2022)
- RNA-seq: 19 SCZ, 18 controls
  - PRJNA695206, Schizophrenia Bulletin Open (2021)

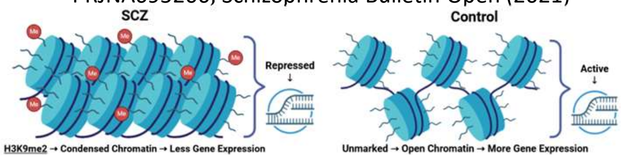
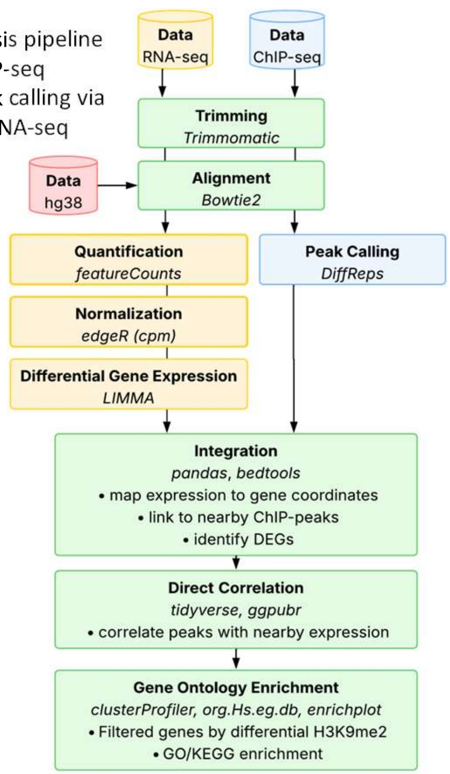


Figure 1: Differences in effects of histone modifications

Figure 2: Analysis pipeline integrating ChIP-seq (H3K9me2 peak calling via DiffReps) and RNA-seq (differential expression via LIMMA) to identify genes with epigenetic and transcriptional changes in schizophrenia



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Effects of **H3K9me2** on **Gene Expression** in **Schizophrenia**

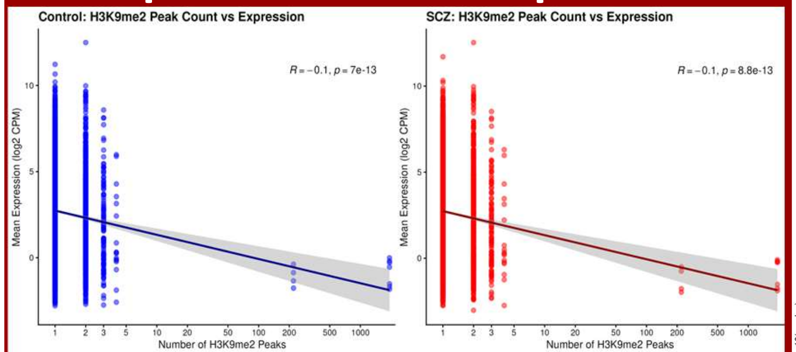

Figure 3: Scatter plot of H3K9me2 peak count versus mean gene expression for control cases (left) and a schizophrenia cases (right) from one sample

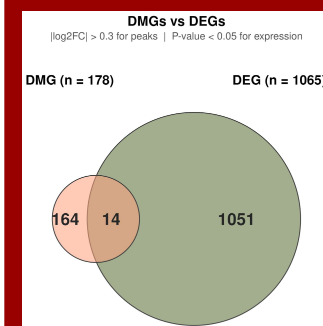

Figure 4: Venn diagram of differentially modified genes (DMGs; H3K9me2 peaks, |log2FC| > 0.3) and differentially expressed genes (DEGs; raw p < 0.05) in SCZ vs control. 14 genes show both differential H3K9me2 modification and differential expression, representing candidate genes for epigenetic regulation in schizophrenia

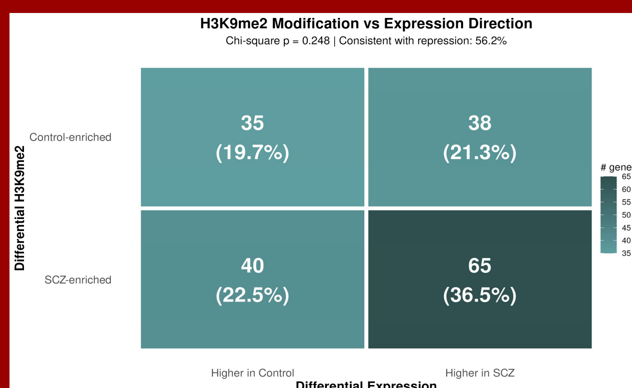

Figure 5. Confusion Matrix comparing direction of differential H3K9me2 modification (SCZ-enriched vs Control-enriched) against direction of differential expression (higher in SCZ vs higher in Control)

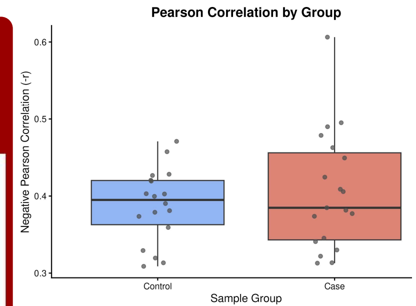

Figure 6: Box plot showing negative Pearson correlation values between H3K9me2 signal and expression of each peak's nearest gene grouped by sample type from all samples

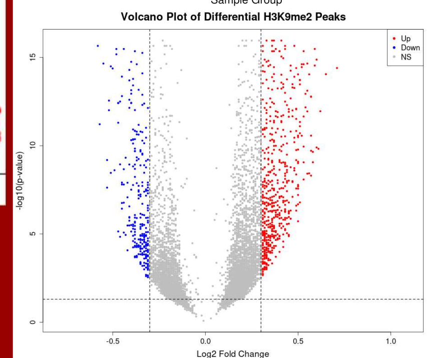

Figure 7: Volcano plot of differential H3K9me2 peaks between SCZ and control samples. Peaks with a significant increase of signal in SCZ are shown in red (Up), decreased in blue (Down), and non-significant in gray (NS)

Discussion

- Control-enriched H3K9me2 peaks showed negative correlation with expression, consistent with repression
- SCZ-enriched peaks showed weak positive correlation with expression change suggesting dysregulated epigenetic function
- Direction of H3K9me2 change at individual genes did not predict direction of expression change
- GO of overlapping DMGs/DEGs yielded no significant results

Conclusion

- H3K9me2 acts as a repressive mark in controls but shows altered function in SCZ where increased H3K9me2 correlates with gene upregulation rather than repression
- SCZ brains show differential H3K9me2 modification (both gains and losses), but this does not translate to predictable per-gene expression changes
- This suggests H3K9me2 operates at the chromatin domain level rather than as a precise per-gene switch

Future Directions

- Create a co-modification network to identify relationships between genes that are modified simultaneously
- Perform GSEA to identify H3K9me2 enriched pathways

Limitations

- Datasets came from different cohorts & brain regions
- H3K9me2 acts over broad heterochromatin domains
- Many other factors regulate expression